

Robust and Comprehensive Diagnostics for Train-Test Data Splitting: Supplementary Materials

Mohammed Mundher Thakir
University of Anbar, Ramadi, Iraq

May 8, 2026

1 Extended Simulation Results

1.1 Type I Error Evaluation

Table 1: Type I error evaluation ($\alpha = 0.05$, $n = 500$).

Measure	Empirical Rejection Rate	95% CI	Calibration
Robust MDAS (Λ_R)	0.048	[0.034, 0.062]	Yes
Energy Distance	0.051	[0.037, 0.065]	Yes
MMD	0.046	[0.032, 0.060]	Yes
KS Battery	0.042	[0.029, 0.055]	Yes
Combined	0.052	[0.038, 0.066]	Yes

1.2 Sensitivity Analysis

Table 2: Detection of lactate distribution shift.

Shift Magnitude (SD)	P-value	Decision
0.0	0.65	ADEQUATE
0.5	0.15	ADEQUATE
0.7	0.04	INADEQUATE
1.0	0.00	INADEQUATE

1.3 Outlier Robustness

Table 3: Effect of outliers on classical and robust MDAS.

Outliers	Classical MDAS	Robust MDAS	Robust Decision
0	3.42	3.39	ADEQUATE
1	6.87	3.44	ADEQUATE
5	14.23	3.61	ADEQUATE
10	∞	3.78	ADEQUATE

2 Code Availability

R package: <https://github.com/Mado-ani/robustMDAS>